

Math 147

Analysis

Qinghao Hu

September 21, 2026

Contents

1	Foundations of Calculus	3
1.1	Lecture 1	3
1.1.1	What Is Calculus?	3
1.1.2	Numbers	3
1.1.3	Statements and Quantifiers	4
1.2	Mathematical Induction	5
1.2.1	Well-Ordering Principle	6
1.3	The Algebraic and Order Properties of $\mathbb{R}$	7
1.3.1	The Absolute Value	10
1.4	Bounds	12
1.5	The Completeness Property of $\mathbb{R}$	14
1.5.1	The Archimedean Property	15
1.5.2	Dense of $\mathbb{Q}$	16
1.6	Limits	18

1 Foundations of Calculus

1.1 Lecture 1

1.1.1 What Is Calculus?

Differential and integral calculus are two branches of calculus.

Differential calculus: The study of slopes and tangent lines.

Integral calculus: The study of areas and accumulation.

1.1.2 Numbers

The number systems discussed in this lecture include

$$\begin{aligned}\mathbb{N} &= \{1, 2, 3, \dots\}, \\ \mathbb{Z} &= \{\dots, -1, 0, 1, \dots\}, \\ \mathbb{Q} &= \left\{ \frac{m}{n} : m, n \in \mathbb{Z}, n \neq 0 \right\}.\end{aligned}$$

The natural numbers are closed under addition. The integers are closed under addition, subtraction, and multiplication, but they are not closed under division; for example, $1/2 \notin \mathbb{Z}$. Therefore, we introduce the rational numbers to include quotients of integers with nonzero denominators.

Theorem 1.1.1 (Irrationality of $\sqrt{2}$)

The number $\sqrt{2}$ is irrational; equivalently, $\sqrt{2} \notin \mathbb{Q}$.

Proof

Suppose, for contradiction, that $\sqrt{2} \in \mathbb{Q}$. Then we can write

$$\sqrt{2} = \frac{m}{n}, \quad m, n \in \mathbb{Z}, \quad n > 0, \quad \gcd(m, n) = 1.$$

Squaring both sides gives

$$2 = \frac{m^2}{n^2} \implies m^2 = 2n^2.$$

Thus m^2 is even. An odd integer has an odd square, so m must be even. Write $m = 2\ell$ for some $\ell \in \mathbb{Z}$. Substituting this into the equation above gives

$$\begin{aligned}2n^2 &= 4\ell^2, \\ n^2 &= 2\ell^2.\end{aligned}$$

Therefore n^2 is even, and the same parity argument shows that n is even.

This means that both m and n are even, contradicting $\gcd(m, n) = 1$. Therefore, $\sqrt{2} \notin \mathbb{Q}$. ■

The irrational numbers are the real numbers that are not rational:

$$\mathbb{R} \setminus \mathbb{Q} = \{x \in \mathbb{R} : x \notin \mathbb{Q}\}.$$

Example 1.1.2 (Decimal representation)

Every nonnegative real number has a decimal representation of the form

$$x = a + \sum_{i=1}^{\infty} \frac{x_i}{10^i}, \quad a \in \mathbb{N} \cup \{0\}, \quad x_i \in \{0, 1, 2, \dots, 9\}.$$

Negative real numbers are the negatives of such expansions. Some numbers have two decimal representations, for example $1.999\dots = 2.000\dots$.

Example 1.1.3 (Recurring decimal)

Is the statement $1.999\dots = 2$ correct?

Yes. Let $x = 1.999\dots$. Then

$$\begin{aligned} x &= 1.999\dots, \\ 10x &= 19.999\dots, \\ 9x &= 18, \\ x &= 2. \end{aligned}$$

1.1.3 Statements and Quantifiers**Definition 1.1.4** (Statement)

A statement is a sentence that has a definite truth value: it is either true or false.

The **universal quantifier** $\forall$ means “for all.” For example,

Example 1.1.5

The statement

$$\forall x \in \mathbb{R}, \quad x^2 - x \geq 0$$

is false; for instance, it fails when $x = \frac{1}{2}$.

The **existential quantifier** $\exists$ means “there exists.” For example,

Example 1.1.6

The statement

$$\exists x \in \mathbb{R}, \quad x^2 - x \geq 0$$

is true; for instance, $x = 2$ satisfies the inequality.

Proposition 1.1.7

For every $x \in \mathbb{R}$, there exists $y \in \mathbb{R}$ such that

$$x^3 - y^3 = 1.$$

Proof

Let $x \in \mathbb{R}$ and take $y = \sqrt[3]{x^3 - 1}$. Then

$$x^3 - y^3 = x^3 - (x^3 - 1) = 1.$$

■

Proposition 1.1.8

The statement

$$\exists y \in \mathbb{R}, \quad \forall x \in \mathbb{R}, \quad x^3 - y^3 = 1$$

is false.

Its negation is

$$\forall y \in \mathbb{R}, \quad \exists x \in \mathbb{R}, \quad x^3 - y^3 \neq 1.$$

To prove the negation, let $y \in \mathbb{R}$ and take $x = y$. Then $x^3 - y^3 = 0 \neq 1$.

Remark

The order of quantifiers matters.

1.2 Mathematical Induction**Definition 1.2.1** (Principle of Mathematical Induction)

Let $P(n)$ be a statement depending on $n \in \mathbb{N}$. If the following two statements are true:

- (1) $P(1)$ is true;
- (2) for every $k \in \mathbb{N}$, if $P(k)$ is true, then $P(k+1)$ is true;

then

$$\forall n \in \mathbb{N}, \quad P(n)$$

is true.

Here is an example:

Proposition 1.2.2 (Sum of the First n Natural Numbers)

For every $n \in \mathbb{N}$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof

Define $P(n)$ to be the statement

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

The base case is $n = 1$:

$$1 = \frac{1(1+1)}{2}.$$

For the inductive step, let $k \in \mathbb{N}$ and assume that $P(k)$ is true. Then

$$\begin{aligned} 1 + 2 + \cdots + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

Thus, $P(k+1)$ is true. By the principle of mathematical induction, the proposition follows. ■

The principle of mathematical induction has the following equivalent set formulation.

Proposition 1.2.3

Let $S \subseteq \mathbb{N}$. If

- (i) $1 \in S$;
- (ii) for every $k \in S$, $k+1 \in S$;

then $S = \mathbb{N}$.

Proof

Define $P(n)$ to be the statement $n \in S$. The two assumptions give the base case $P(1)$ and the inductive implication $P(k) \Rightarrow P(k+1)$. Hence, by mathematical induction, every $n \in \mathbb{N}$ belongs to S . Since $S \subseteq \mathbb{N}$, it follows that $S = \mathbb{N}$. ■

1.2.1 Well-Ordering Principle

Theorem 1.2.4 (Well-Ordering Principle)

Every nonempty subset S of $\mathbb{N}$ has a smallest element.

Proof

Let $S \subseteq \mathbb{N}$ be nonempty. Suppose, for contradiction, that S has no smallest element. Define

$$T = \{n \in \mathbb{N} : \{1, 2, \dots, n\} \cap S = \emptyset\}.$$

Since S has no smallest element, $1 \notin S$, so $1 \in T$. Suppose $k \in T$. If $k+1 \notin T$, then $k+1 \in S$, while no element of $\{1, 2, \dots, k\}$ belongs to S . Thus, $k+1$ would be the smallest element of S , a contradiction. Therefore, $k+1 \in T$.

By mathematical induction, $T = \mathbb{N}$. This implies that no natural number belongs to S , so

$S = \emptyset$, contradicting the assumption that S is nonempty. Therefore, S has a smallest element. ■

Proposition 1.2.5

Using the well-ordering principle, prove that for every $n \in \mathbb{N}$,

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Proof

Let $P(n)$ denote the statement

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Suppose, for contradiction, that $P(n)$ fails for some n ; that is,

$$\neg(\forall n \in \mathbb{N}, P(n)) \quad \text{equivalently} \quad \exists n \in \mathbb{N}, \neg P(n).$$

Then the set

$$S := \{n \in \mathbb{N} : \neg P(n)\}$$

is nonempty. By the well-ordering principle, S has a smallest element k . Since $P(1)$ holds, $1 \notin S$, so $k > 1$ and hence $k-1 \in \mathbb{N}$.

By minimality of k , we have $k-1 \notin S$, so $P(k-1)$ is true:

$$1^2 + 2^2 + \cdots + (k-1)^2 = \frac{(k-1)k(2(k-1)+1)}{6} = \frac{(k-1)k(2k-1)}{6}.$$

Adding k^2 to both sides gives

$$\begin{aligned} 1^2 + \cdots + (k-1)^2 + k^2 &= \frac{(k-1)k(2k-1)}{6} + k^2 \\ &= \frac{(k-1)k(2k-1) + 6k^2}{6} \\ &= \frac{k((k-1)(2k-1) + 6k)}{6} \\ &= \frac{k(2k^2 + 3k + 1)}{6} = \frac{k(k+1)(2k+1)}{6}, \end{aligned}$$

so $P(k)$ is in fact true. This contradicts $k \in S$. Therefore $S = \emptyset$, and $P(n)$ holds for every $n \in \mathbb{N}$. ■

1.3 The Algebraic and Order Properties of $\mathbb{R}$

The real numbers $\mathbb{R}$ form a *complete ordered field*. We build up this description in three stages: the field (algebraic) axioms, the order axioms, and the completeness axiom.

Definition 1.3.1 (Field)

A **field** is a set F equipped with two binary operations

$$+ : F \times F \rightarrow F, \quad \cdot : F \times F \rightarrow F,$$

satisfying the following axioms for all $a, b, c \in F$.

F1. Associativity of addition.

$$a + (b + c) = (a + b) + c.$$

F2. Commutativity of addition.

$$a + b = b + a.$$

F3. Additive identity. There exists an element $0 \in F$ such that

$$a + 0 = a.$$

F4. Additive inverses. For every $a \in F$, there exists an element $-a \in F$ such that

$$a + (-a) = 0.$$

F5. Associativity of multiplication.

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c.$$

F6. Commutativity of multiplication.

$$a \cdot b = b \cdot a.$$

F7. Multiplicative identity. There exists an element $1 \in F$, with $1 \neq 0$, such that

$$a \cdot 1 = a.$$

F8. Multiplicative inverses. For every $a \in F$ with $a \neq 0$, there exists an element $a^{-1} \in F$ such that

$$a \cdot a^{-1} = 1.$$

F9. Distributive law.

$$a \cdot (b + c) = a \cdot b + a \cdot c.$$

Remark

The rational numbers $\mathbb{Q}$ also form a field.

Theorem 1.3.2 a) If $z, a \in \mathbb{R}$ with $z + a = a$, then $z = 0$.

b) If $u \in \mathbb{R}$ and $b \neq 0$ with $u \cdot b = b$, then $u = 1$.

c) If $a \in \mathbb{R}$, then $a \cdot 0 = 0$.

Proof

a)

$$z = z + 0 = z + (a - a) = (z + a) + (-a) = a + (-a) = 0.$$

b) Multiplying $u \cdot b = b$ by b^{-1} ,

$$u = u \cdot 1 = u \cdot (b \cdot b^{-1}) = (u \cdot b) \cdot b^{-1} = b \cdot b^{-1} = 1.$$

c)

$$a \cdot 0 = a \cdot (0 + 0) = a \cdot 0 + a \cdot 0.$$

Adding $-(a \cdot 0)$ to both sides, and using associativity, gives $0 = a \cdot 0$.

■

Definition 1.3.3 (The Order Properties of $\mathbb{R}$)

There is a nonempty subset P of $\mathbb{R}$, called the set of **positive real numbers**, that satisfies the following properties.

O1. Trichotomy law. For every $a \in \mathbb{R}$, exactly one of the following holds:

$$a = 0, \quad a \in P, \quad -a \in P.$$

O2. Closure under addition. If $a, b \in P$, then $a + b \in P$.

O3. Closure under multiplication. If $a, b \in P$, then $a \cdot b \in P$.

Definition 1.3.4 (Order relation)

Let $a, b \in \mathbb{R}$.

(i) If $a - b \in P$, then we write $a > b$ or $b < a$.

(ii) If $a - b \in P \cup \{0\}$, then we write $a \geq b$ or $b \leq a$.

Remark

$\mathbb{Q}$ is also ordered, so we are still missing something: completeness.

Definition 1.3.5 (Completeness Axiom)

If $A \subseteq \mathbb{R}$ is nonempty and bounded above, then A has a least upper bound in $\mathbb{R}$.

Example 1.3.6

Prove that if $a \in \mathbb{R}$ and $a \neq 0$, then $a^2 > 0$.

Proof

Since $a \neq 0$, the trichotomy law gives two cases.

Case 1: $a \in P$. By closure under multiplication,

$$a^2 = a \cdot a \in P, \quad \text{so } a^2 > 0.$$

Case 2: $a \notin P$. Since $a \neq 0$, trichotomy gives $-a \in P$, so by closure under multiplication $(-a)(-a) \in P$. It remains to show that $a^2 = (-a)(-a)$:

$$\begin{aligned} a + (-a) = 0 &\implies (-a)(a + (-a)) = (-a) \cdot 0 = 0 \\ &\implies (-a)(a) + (-a)(-a) = 0 \\ &\implies (-a)(-a) = -((-a)a) = -(-a^2) = a^2. \end{aligned}$$

Hence $a^2 = (-a)(-a) \in P$, so $a^2 > 0$. In either case $a^2 > 0$. ■

1.3.1 The Absolute Value

Definition 1.3.7

Given $a \in \mathbb{R}$, we define the **absolute value** $|a|$ by

$$|a| = \begin{cases} a, & a \geq 0, \\ -a, & a < 0. \end{cases}$$

Theorem 1.3.8 (a) $|ab| = |a||b|$ for all $a, b \in \mathbb{R}$.

(b) $|a|^2 = a^2$ for all $a \in \mathbb{R}$.

(c) For $a \in \mathbb{R}$ and $c > 0$,

$$|a| \leq c \iff -c \leq a \leq c.$$

(d) For every $a \in \mathbb{R}$,

$$-|a| \leq a \leq |a|.$$

Proof

The definition immediately gives $|-a| = |a|$: if $a \geq 0$, then $-a \leq 0$, and if $a < 0$, then $-a > 0$. It also gives

$$-|a| \leq a \leq |a|,$$

proving (d). In the two cases $a \geq 0$ and $a < 0$, respectively, $|a|^2 = a^2$ is immediate; this proves (b).

For (a), there are four sign combinations. If $a, b \geq 0$, then $ab \geq 0$, so $|ab| = ab = |a||b|$. If $a \geq 0 > b$, then $ab \leq 0$, so $|ab| = -ab = a(-b) = |a||b|$. The case $b \geq 0 > a$ is the same with the roles reversed. Finally, if $a, b < 0$, then $ab > 0$, and

$$|ab| = ab = (-a)(-b) = |a||b|.$$

Finally let $c > 0$. Suppose first that $|a| \leq c$. If $a \geq 0$, then $|a| = a$, so $a \leq c$; also $-c < 0 \leq a$.

If $a < 0$, then $|a| = -a$, so $-a \leq c$, equivalently $-c \leq a$; also $a < 0 < c$. Hence $-c \leq a \leq c$ in either case.

Conversely, suppose $-c \leq a \leq c$. If $a \geq 0$, then $|a| = a \leq c$. If $a < 0$, multiplying $-c \leq a$ by -1 reverses the inequality and gives $-a \leq c$, hence $|a| \leq c$. This proves (c). ■

Proposition 1.3.9 (Triangle Inequality)

If $a, b \in \mathbb{R}$, then $|a + b| \leq |a| + |b|$.

Proof

Let $a, b \in \mathbb{R}$. We know that

$$\begin{aligned} -|a| &\leq a \leq |a| \\ -|b| &\leq b \leq |b| \end{aligned}$$

Thus,

$$\begin{aligned} -|a| - |b| &\leq a + b \leq |a| + |b| \\ \iff -(|a| + |b|) &\leq a + b \leq |a| + |b| \end{aligned}$$

If $|a| + |b| = 0$, then the displayed bounds give $a + b = 0$, so the claim is immediate. Otherwise, part (c) of the preceding theorem gives

$$|a + b| \leq |a| + |b|.$$

■

Corollary 1.3.10 (i) $||a| - |b|| \leq |a - b|$.

(ii) $|a - b| \leq |a| + |b|$

Proof of (1)

Let $a, b \in \mathbb{R}$. We know

$$\begin{aligned} a &= a - b + b \implies |a| = |(a - b) + b| \leq |a - b| + |b| \\ &\implies |a| - |b| \leq |a - b| \end{aligned}$$

$$\begin{aligned} b &= b - a + a \implies |b| = |(b - a) + a| \leq |b - a| + |a| \\ &\implies |b| - |a| \leq |b - a| = |-(a - b)| = |a - b|. \\ &\implies |a| - |b| \geq -|a - b| \end{aligned}$$

Thus,

$$-|a - b| \leq |a| - |b| \leq |a - b|$$

Therefore,

$$||a| - |b|| \leq |a - b|$$

■

Proof of (2)

Let $a, b \in \mathbb{R}$. We know

$$|a - b| = |a + (-b)| \leq |a| + |-b| = |a| + |b|.$$

■

1.4 Bounds

Bounds make sense in any ordered set. The special feature of $\mathbb{R}$ is that every nonempty set bounded above has a supremum in $\mathbb{R}$; this need not hold in $\mathbb{Q}$.

Definition 1.4.1 (Bounded Sets)

Let S be a nonempty subset of $\mathbb{R}$. The set S is said to be

- (i) bounded above: there exists $u \in \mathbb{R}$ such that $x \leq u$ for all $x \in S$.
- (ii) bounded below: there exists $w \in \mathbb{R}$ such that $w \leq x$ for all $x \in S$.
- (iii) bounded: it is both bounded above and below.
- (iv) unbounded: not bounded.

If there exists an upper bound u_0 of S such that whenever u is an upper bound for S we have $u_0 \leq u$, then u_0 is called the least upper bound, or the supremum, of S .

$$u_0 := \sup S$$

If there exists a lower bound w_0 of S such that whenever w is a lower bound for S we have $w_0 \geq w$, then w_0 is called the greatest lower bound, or the infimum, of S .

$$w_0 := \inf S$$

Remark

There can be only one supremum/infimum of a given set S of $\mathbb{R}$.

Idea: Suppose u_1 and u_2 are both suprema of S .

Since u_1 is a supremum and u_2 is an upper bound, we get

$$u_1 \leq u_2.$$

Since u_2 is a supremum and u_1 is an upper bound, we get

$$u_2 \leq u_1.$$

Thus, $u_1 = u_2$.

Lemma 1.4.2

An upper bound u of a nonempty set S in $\mathbb{R}$ is the supremum of S if and only if

$$(\forall \epsilon > 0)(\exists x_\epsilon \in S)u - \epsilon < x_\epsilon.$$

Proof

First suppose $u = \sup S$. Let $\epsilon > 0$. Since

$$u - \epsilon < u,$$

the number $u - \epsilon$ cannot be an upper bound: otherwise it would be an upper bound smaller than u . By the definition of upper bound, there is therefore some $x_\epsilon \in S$ such that

$$u - \epsilon < x_\epsilon \leq u.$$

Conversely, suppose u is an upper bound and that for every $\epsilon > 0$ there exists $x_\epsilon \in S$ with $u - \epsilon < x_\epsilon$. If u were not the least upper bound, there would be another upper bound $w < u$. Choose

$$\epsilon = u - w > 0.$$

The assumed condition then yields

$$w = u - \epsilon < x_\epsilon,$$

contradicting that w is an upper bound. Hence $u = \sup S$. ■

Example 1.4.3

Consider the set $S = \{x \in \mathbb{R} : 0 \leq x \leq 1\}$.

– $\sup S = 1$.

Proof

Every $x \in S$ satisfies $x \leq 1$, so 1 is an upper bound. To see directly that no smaller number works, let $u < 1$. If $u < 0$, then $0 \in S$ lies above u . If $0 \leq u < 1$, set $x = (u + 1)/2$. Then

$$u < x = \frac{u + 1}{2} < 1,$$

so $x \in S$ lies above u . Thus every number below 1 fails to be an upper bound, and $\sup S = 1$. ■

– $\inf S = 0$.

Proof

Since $x \geq 0$ for all $x \in S$, 0 is a lower bound. If $w > 0$, then $0 \in S$ satisfies $0 < w$, so w is not a lower bound. Thus no number greater than 0 is a lower bound, and $\inf S = 0$. ■

1.5 The Completeness Property of $\mathbb{R}$

Axiom (Completeness Property)

Every nonempty set of real numbers that has an upper bound also has a supremum in $\mathbb{R}$.

Corollary 1.5.1

Every nonempty set of real numbers that has a lower bound also has an infimum in $\mathbb{R}$.

Proof

Let $S \subseteq \mathbb{R}$ be nonempty with a lower bound w , that is, $w \leq s$ for all $s \in S$. As a consequence, $-w \geq -s$ for all $s \in S$.

Define $S' = \{-s : s \in S\}$. Then S' is nonempty, and

$$-w \geq -s, \forall s \in S \implies -w \text{ is an upper bound for } S'.$$

By the Completeness Axiom, S' has a supremum. Say

$$u := \sup S' \in \mathbb{R}.$$

We claim that $-u = \inf S$.

- **Lower bound:** Since $u = \sup S' \geq -s$ for all $s \in S$, we get $-u \leq s$ for all $s \in S$. So $-u$ is a lower bound for S .
- **Greatest lower bound:** Let w be any lower bound for S . As shown above, $-w$ is then an upper bound for S' , so $u = \sup S' \leq -w$, which gives $w \leq -u$.

Thus $-u$ is a lower bound for S that is at least as large as every other lower bound, so $-u = \inf S$. ■

Example 1.5.2

$$S = \left\{ \left(1 + \frac{1}{n}\right)^n : n \in \mathbb{N} \right\}$$

For every $n \in \mathbb{N}$, the binomial theorem gives

$$\left(1 + \frac{1}{n}\right)^n = 1 + \binom{n}{1} \frac{1}{n} + \sum_{k=2}^n \binom{n}{k} \frac{1}{n^k} > 1.$$

Indeed, for $0 \leq k \leq n$,

$$\binom{n}{k} = \frac{n(n-1) \cdots (n-k+1)}{k!} \leq \frac{n^k}{k!}.$$

For $k \geq 2$, we have $k! \geq 2^{k-1}$, because each factor from 2 through k is at least 2. Therefore

$$\left(1 + \frac{1}{n}\right)^n \leq \sum_{k=0}^n \frac{1}{k!} < 1 + 1 + \sum_{k=2}^{\infty} \frac{1}{2^{k-1}} = 3.$$

| Thus $1 < (1 + 1/n)^n < 3$ for all $n \in \mathbb{N}$.

1.5.1 The Archimedean Property

We now prove that $\mathbb{N}$ is not bounded above in $\mathbb{R}$. The proof combines two ingredients:

- Completeness of $\mathbb{R}$.
- The inductive property of $\mathbb{N}$ (that $\mathbb{N}$ is closed under $+1$).

Definition 1.5.3 (Archimedean Property)

If $x \in \mathbb{R}$, then there exists $n_x \in \mathbb{N}$ such that $x \leq n_x$.

Proof

Suppose, for contradiction, that $\exists x \in \mathbb{R}$ such that $x > n$ for all $n \in \mathbb{N}$.

Then x is an upper bound for $\mathbb{N}$. Since $1 \in \mathbb{N}$, $\mathbb{N} \neq \emptyset$, so by the Completeness Axiom, $\mathbb{N}$ has a supremum in $\mathbb{R}$.

Say $u = \sup \mathbb{N}$. Since

$$u - 1 < u,$$

$u - 1$ is not an upper bound for $\mathbb{N}$. Hence

$$\exists m \in \mathbb{N} \text{ such that } u - 1 < m,$$

which gives

$$u < m + 1.$$

But $m + 1 \in \mathbb{N}$ (since $\mathbb{N}$ is closed under $+1$), so $m + 1 \leq u$, contradicting $u < m + 1$.

Therefore no such x exists; that is, for every $x \in \mathbb{R}$ there exists $n_x \in \mathbb{N}$ with $x \leq n_x$. ■

Proposition 1.5.4

There exists a real number x such that $x^2 = 2$.

Proof

Define

$$S = \{t \in \mathbb{R} : t \geq 0, t^2 < 2\}.$$

We have $1 \in S$, so $S \neq \emptyset$, and 2 is an upper bound for S : if $t \in S$ and $t > 2$, then $t^2 > 4 > 2$, a contradiction. By the Completeness Axiom, S has a supremum in $\mathbb{R}$.

Let $x = \sup S$. Note $x \geq 1 > 0$. We show $x^2 = 2$ by ruling out $x^2 < 2$ and $x^2 > 2$.

Case 1: $x^2 < 2$. We find $n \in \mathbb{N}$ large enough that $x + \frac{1}{n} \in S$, contradicting that x is an upper bound for S . Since $x \geq 1$ and $n \geq 1$,

$$\left(x + \frac{1}{n}\right)^2 = x^2 + \frac{2x}{n} + \frac{1}{n^2} \leq x^2 + \frac{2x}{n} + \frac{1}{n} = x^2 + \frac{2x+1}{n}.$$

Since $x^2 < 2$, the Archimedean Property lets us choose $n \in \mathbb{N}$ large enough that

$$\frac{2x+1}{n} < 2 - x^2.$$

Then

$$\left(x + \frac{1}{n}\right)^2 < x^2 + (2 - x^2) = 2,$$

so $x + \frac{1}{n} \in S$. But $x + \frac{1}{n} > x = \sup S$, a contradiction. Hence $x^2 \geq 2$.

Case 2: $x^2 > 2$. We find $n \in \mathbb{N}$ large enough that $x - \frac{1}{n}$ is an upper bound for S smaller than x , contradicting that $x = \sup S$. Since

$$\left(x - \frac{1}{n}\right)^2 = x^2 - \frac{2x}{n} + \frac{1}{n^2} > x^2 - \frac{2x}{n},$$

the Archimedean Property lets us choose $n \in \mathbb{N}$ large enough that

$$\frac{2x}{n} < x^2 - 2.$$

Then

$$\left(x - \frac{1}{n}\right)^2 > x^2 - (x^2 - 2) = 2.$$

So for any $t \in S$, we have $t^2 < 2 < \left(x - \frac{1}{n}\right)^2$. Also $x - 1/n \geq 0$ because $x \geq 1$ and $n \geq 1$. Since $u \mapsto u^2$ is strictly increasing on $[0, \infty)$, this implies $t < x - \frac{1}{n}$. Thus $x - \frac{1}{n}$ is an upper bound for S smaller than $x = \sup S$, a contradiction. Hence $x^2 \leq 2$.

Combining both cases, $x^2 = 2$. ■

1.5.2 Dense of $\mathbb{Q}$

Remark

Consider the set

$$T = \{r \in \mathbb{Q} : r^2 < 2\}.$$

Then $T \neq \emptyset$, and T is bounded above in $\mathbb{Q}$. For example,

$$0 \in T \quad \text{and} \quad r < 2 \quad \text{for every } r \in T.$$

If T is regarded as a subset of $\mathbb{R}$, then by the completeness of $\mathbb{R}$, the supremum

$$y = \sup_{\mathbb{R}} T$$

exists.

We can prove that

$$y^2 = 2.$$

Hence

$$y = \sqrt{2}.$$

Since

$$\sqrt{2} \notin \mathbb{Q},$$

it follows that T has no supremum in $\mathbb{Q}$.

Therefore, there exists a nonempty subset $T \subset \mathbb{Q}$ which is bounded above in $\mathbb{Q}$, but which has no least upper bound in $\mathbb{Q}$. Thus $\mathbb{Q}$ is not complete.

Theorem 1.5.5

If x and y are any real numbers with $x < y$, then there exists a rational number $r \in \mathbb{Q}$ such that $x < r < y$.

Proof

Suppose x and y are arbitrary real numbers where $x < y$. We have

$$x < y \iff y - x > 0.$$

According to the Archimedean Property, it follows that

$$(\exists n \in \mathbb{N}) \frac{1}{n} < y - x. \iff yn - xn > 1 \iff yn > 1 + xn.$$

■

Corollary 1.5.6

If x and y are any real numbers with $x < y$, then there exists an irrational number z such that $x < z < y$.

Corollary 1.5.7

Let $x, y \in \mathbb{R}$, with $x < y$. Then there exists infinitely many rationals between x and y .

Are there more rational or irrational numbers? To answer this question, we need to know in what sense.

Definition 1.5.8 (Cardinality)

Size of the set.

Definition 1.5.9

Two sets have the same cardinality if there is a 1 - 1 (injective) and onto (surjective) function between sets.

Example 1.5.10 (a) $\mathbb{N}$ and the set of even natural numbers.

(b) $\mathbb{N}$ and $\mathbb{Z}$.

Definition 1.5.11

S is called countable if S has the same cardinality as $\mathbb{N}$. Otherwise, S is called uncountable.

1.6 Limits

A sequence of real numbers in a function defined on the set $N = \{1, 2, 3, \dots\}$ of natural numbers whose range is contained in the set $\mathbb{R}$ of real numbers.

$$X : \mathbb{N} \rightarrow \mathbb{R}$$

x_n : element of the sequence.

Notation: $\{x_n\}, \{x_n\}_{n=1}^{\infty}$

Definition 1.6.1

A sequence $\{x_n\}$ in $\mathbb{R}$ is said to converge to $a \in \mathbb{R}$, or a is said to be a limit of $\{x_n\}$, if for every $\epsilon > 0$ there exists a natural number $n_\epsilon \in \mathbb{N}$ such that for all $n \geq n_\epsilon$ the term x_n satisfy

$$|x_n - a| < \epsilon.$$

The notation is

$$\lim_{n \rightarrow \infty} x_n = a.$$